Free Will and Framework Inertia

A Semi-Formal Treatment in Textbook Style

LLM Mock Notes

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Abstract

We propose *Framework Inertia Theory*, a lightweight formal model that explains why people often fail to initiate beneficial activities despite knowing they should. By modelling behavioural states as potential wells and volitional energy as a depletable resource, we derive that free will achieves maximal leverage at *decision windows*—brief moments when inertia is locally minimal—and that *planning* is the principal mechanism for lowering activation barriers in advance. The treatment is deliberately semi-formal: definitions and theorems provide structure, while discussions and examples supply intuition.

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1 Introduction

We frequently encounter the following paradox in everyday life: we *know* we should study, exercise, or pursue a long-term beneficial activity, yet we find ourselves unable to initiate the transition. It is as though an invisible vortex holds us in our current state—scrolling on a phone, lying idle, daydreaming—while rationality screams "move!" but the body and emotions remain leaden.

This is not mere "laziness." It is a phenomenon amenable to structured analysis. In what follows, we develop a small formal framework—*Framework Inertia Theory*—to characterize the problem, locate where free will exerts maximal leverage, and derive why planning is not optional but *necessary*.

2 Framework Inertia

Definition 2.1 (Behavioural State)**.** A **behavioural state** $s \in \mathcal{S}$ is a sustained pattern of activity together with its associated physical environment, cognitive context, and neurochemical reward profile.

Example

Lying in bed watching short videos constitutes a behavioural state s_0 characterised by: (i) a supine physical posture, (ii) a minimal-effort cognitive context, and (iii) a steady stream of dopaminergic micro-rewards. Sitting at a desk studying constitutes a different state s_1 with high cognitive load and delayed reward.

Definition 2.2 (Framework Inertia)**.** The **framework inertia** $I(s)$ of a behavioural state s is the resistance encountered when attempting to transition away from s. Formally,

$$I(s) \;=\; \alpha\, R(s) \;+\; \beta\, C(s) \;+\; \gamma\, E(s),$$

where

- $R(s) \geq 0$ is the *instantaneous reward rate* of the current activity (neurological component),
- $C(s) \geq 0$ is the *cognitive switching cost*—the mental overhead of loading a new context (cognitive component),
- $E(s) \geq 0$ is the *environmental reinforcement*—the degree to which the physical surroundings sustain s (environmental component),

and $\alpha, \beta, \gamma > 0$ are individual weighting constants.

Remark 2.3. The three components of inertia correspond to observable mechanisms:

1. **Neurological:** The current activity continuously releases dopamine; switching to a delayed-reward activity causes a sharp, aversive drop in reward signal.
2. **Cognitive:** Switching demands reloading context ("Where did I leave off?" "What should I start with?"), a cost that the fatigued brain amplifies.
3. **Environmental:** Posture, location, and surrounding objects all reinforce the current state. A person lying in bed inhabits a physical frame that tells the brain "you are resting."

2.1 The Inertia Vortex Model

Definition 2.4 (Potential Well Model)**.** Model the state space $\mathcal{S}$ as a landscape equipped with a potential function $V : \mathcal{S} \to \mathbb{R}$. Each behavioural state s sits at a local minimum (a **potential**

well) of depth proportional to $I(s)$. Transitioning from state s_0 to state s_1 requires surmounting the **activation barrier**

$$\Delta V(s_0 \to s_1) \;=\; V_{\max}(s_0, s_1) - V(s_0),$$

where $V_{\max}(s_0, s_1)$ is the potential at the saddle point between the two wells.

Proposition 2.5 (Difficulty of Escape)**.** *The difficulty of escaping a behavioural state s_0 increases with:*

1. *the depth of the current well (high instantaneous reward $R(s_0)$),*
2. *the height of the activation barrier (large switching cost C and startup friction of the target activity),*
3. *the width of the environmental reinforcement basin $E(s_0)$.*

Discussion

Proposition 2.5 follows directly from Definition 2.4: a deeper well means a larger ΔV to overcome, and a wider basin means the agent must sustain effort over a longer transition path. This is why "powering through with pure grit" fails so often—the activation energy exceeds what the agent can supply in a single burst.

3 Free Will and Decision Windows

Definition 3.1 (Volitional Energy)**. Volitional energy** $W(t) \geq 0$ is the time-varying resource an agent expends to override framework inertia. It satisfies:

1. **Finite capacity:** $W(t) \leq W_{\max}$ for all t.
2. **Depletion:** Active exertion against inertia depletes W at rate $\dot{W} = -\lambda\,\Delta V$, where $\lambda > 0$.
3. **Recovery:** W regenerates during rest and sleep.

Remark 3.2. W is analogous to a battery: it charges during periods of low demand and discharges under sustained exertion. The "ego depletion" literature in psychology, while debated in its strong form, captures the phenomenology that self-control is *effortful* and *limited within a session.*

Definition 3.3 (Decision Window)**.** A **decision window** is a time interval during which framework inertia is locally minimised, i.e. $I(s(t))$ is near a local minimum. The *effective leverage* of volitional energy during a decision window is

$$L(t) \;=\; \frac{W(t)}{\Delta V(t)},$$

which is maximised when ΔV is small and W is large.

Theorem 3.4 (Maximal Leverage of Free Will)**.** *Free will exerts its greatest influence at **decision windows**—moments when framework inertia is locally minimal—rather than during sustained resistance against high inertia.*

Discussion

Since $L(t) = W(t)/\Delta V(t)$, the leverage is highest when ΔV is small. Decision windows are precisely the moments when the activation barrier dips:

- **Start of day:** upon waking, the previous behavioural frame is not yet fully loaded; a new frame can be injected at low cost.
- **Activity boundaries:** the instant a task ends, inertia momentarily collapses to near zero—an ideal insertion point for a new trajectory.
- **Environment transitions:** entering a library or arriving at a gym door physically disrupts the old frame, lowering $E(s)$.
- **Emotional perturbations:** a stimulus—reading an inspiring passage, witnessing an event—temporarily shatters the inertial frame.

The optimal strategy is therefore not to *continuously fight inertia* but to *deploy volitional energy at decision windows and then let the new inertia sustain the desired state.* Once you sit at the desk, open the textbook, and write the first line of notes, the new framework inertia begins to form—and now inertia works *for* you.

Key Insight

The optimal strategy for free will is **not** "constantly resist inertia" but rather "*act decisively at the moment inertia is weakest, then let the new inertia carry you forward.*"

4 The Necessity of Planning

Theorem 4.1 (Planning as Pre-committed Trajectory)**.** *A plan constructed during a period of high volitional energy $W(t_{plan}) \approx W_{\max}$ reduces the effective activation barrier ΔV at future decision windows, thereby increasing leverage $L(t)$ at times when W may be depleted.*

Discussion

From the perspective of Framework Inertia Theory, a plan is *not* merely a to-do list. It is a mechanism that operates on four channels:

1. **Reduces cognitive switching cost C.** With a plan in place, the agent need not deliberate "What should I do next?" at the moment of lowest willpower—that question was already answered during a period of peak clarity.
2. **Compresses decision load at the window.** During a precious decision window the agent faces a single binary choice—"execute the plan"—instead of the multi-step chain "What to do → How to do it → Where to start → For how long," each step of which drains W.
3. **Creates external forces.** Written plans, alarms, scheduled commitments, and accountability partners serve as external forces F that contribute additional energy to surmount ΔV, partially substituting for volitional energy.
4. **Establishes an anticipatory frame.** When the brain already knows "what comes next," the environmental and cognitive components of the barrier are pre-lowered—just as road signs reduce the cognitive load of navigation.

Corollary 4.2 (The Unplanned Agent is Not Free)**.** *An agent without a plan is not exercising freedom; rather, the agent is* ***fully governed by framework inertia****. Every instance of "just five more minutes" is a decision made* by inertia*, not by the agent's free will.*

Discussion

This is a direct consequence of Theorem 4.1. Without a pre-committed trajectory, each transition demands the full activation energy ΔV at the moment of action, while volitional energy W may already be depleted. The agent defaults to the path of least resistance, which is precisely the current inertial state.
Genuine free will therefore *requires* planning: willpower without a pre-laid track is like water without a channel—it disperses aimlessly.

Example

Consider an agent who resolves each morning to exercise after work.

- **Without a plan:** At 6 PM the agent faces $\Delta V =$ high (gym bag unpacked, route undecided, workout unplanned) with $W(6\text{PM})$ near its daily minimum. The agent scrolls social media instead.
- **With a plan:** The gym bag was packed the night before, the workout was written down, and a friend is meeting the agent at the gym entrance. The activation barrier ΔV has been drastically lowered by preparatory actions taken during high-W periods.

5 Practical Methodology

Method 1: Exploit Decision Windows

- **Morning override:** Open your plan first; do not reach for your phone.
- **Boundary insertion:** Immediately after completing a task, consult the next item on the plan—exploit the zero-inertia gap.
- **Spatial anchoring:** Assign specific locations to specific activities so that the environment itself becomes a frame trigger.

Method 2: Lower the Activation Barrier

- **Minimise startup friction:** Bookmark the chapter you will read next; lay out exercise clothes the night before.
- **Two-Minute Rule:** Do not think "I will study for two hours." Think "I will look at this for two minutes." The sole objective is to cross the barrier; once across, inertia takes over.

Method 3: Conserve the Volitional Battery

- **Eliminate trivial decisions:** Routinise daily choices (meals, clothing) to preserve W for high-stakes transitions.
- **Front-load demanding tasks:** Schedule the most initiative-intensive work during peak W periods (typically morning).
- **Eliminate rather than resist:** Removing a high-reward distractor from your environment (turning off the phone, leaving the room) is vastly more efficient than sustaining volitional resistance against it.

6 Summary

Concept	Core Meaning
Framework Inertia	The tendency to remain in the current behavioural state; a physical phenomenon, not a moral failing.
Inertia Vortex	The higher the instant reward and switching cost, the harder it is to escape the current state.
Decision Window	A critical point of state transition where free will has maximal leverage.
Nature of Planning	Using lucid-state volition to pre-lay tracks for future depleted-state selves.
Genuine Freedom	Not unplanned impulsivity, but directing willpower precisely at critical junctures via planning.

Key Insight

One-line summary. Do not attempt to pit free will against inertia head-on—instead, use free will to *redirect* inertia at the right moment, then let inertia do the rest.